Optical Pumping of Rubidium

University of Lethbridge
PHYS3925 - Intermediate Experimental Physics

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Abstract

Optical pumping is a process in which light is used to force atoms into a higher energy level. This can be utilized in the study of energy within an atom, which becomes interesting when those energies undergo fine or hyperfine splitting due to an applied magnetic field, known as the Zeeman Effect. The objectives of this lab were to explore the Zeeman effect in linear and quadratic regimes, and observe hyperfine splitting, as well as to become familiar with the physics of optical pumping. Using a rubidium vapour cell and rubidium resonance lamp, optical pumping was used to observe the Zeeman Effect in two isotopes of rubidium, exploring the different energies that result from different quantum numbers. The Landé g-factors of ^{87}Rb and ^{85}Rb were found to be 0.55 ± 0.15 and 0.32 ± 0.04, respectively. The main horizontal coils of the apparatus were found to create a field of 8.59 Gauss/Amp. The quadratic zeeman effect was observed in both isotopes of rubidium, but with some conflicts in the predicted and observed number of resonances.

Introduction and Theory

Optical pumping is a process by which light is used to increase the energy level of an electron, atom, or molecule through photon absorption. The methods by which it occurs and can be studied are best explored after a brief review of atomic physics.

Electrons may have an orbital angular momentum, $\vec{L}$, as well as spin angular momentum, $\vec{S}$ [3]. A total angular momentum, $\vec{J}$ can be defined, where $\vec{J} = \vec{L} + \vec{S}$. Within an atom, the orbital angular momentum dictates an electron's orbital, and its spin angular momentum determines if the electron is in the spin up or spin down states. The energy difference between states of total electron angular momentum based on spin is known as fine structure splitting. When the nucleus of the atom containing an electron is taken into account, a nuclear angular momentum, $\vec{I}$ must be considered. The net angular momentum of the atomic nucleus and a single valence electron is then denoted by $\vec{F}$. The energy difference between states of total atomic angular momentum based on nuclear spin is known as hyperfine structure splitting. Each component of angular momentum is quantized and follows the conservation of momentum, allowing atomic collisions and the transfer of energy and momenta to be studied .

Rubidium is an alkali metal, and therefore has only one valence electron. This structure is ideal to study absorption, as only the single valence electron is available for photon interactions. Atomic behaviour is somewhat simplified, and can be modelled to a reasonable accuracy as a hydrogen-like atom where the pseudo-hydrogen nucleus contains the rubidium nucleus and first 36 electrons. In the case of optical pumping of rubidium, transitions between the ground and first excited states have an energy difference corresponding to radiation with a wavelength of about 795 nm. If these photons are circularly polarized such that they have an angular momentum equal to $\hbar$, absorption of these photons by rubidium will result in both a change of electron energy level and net atomic angular momentum.

The Zeeman Effect refers to a phenomenon where in the presence of an external magnetic field, atomic energy levels split into several components with very small differences in energy. For strong magnetic fields, the difference between these hyperfine levels is given by the Breit-Rabi equation:

$$W = \frac{-\Delta W}{2(2I+1)} + g_I \mu_B B m_F \pm \frac{\Delta W}{2}\sqrt{1 + \frac{2m_F}{2I+1}x + x^2} \tag{1}$$

where

$$x = (g_J - g_I)\frac{\mu_B B}{\Delta W} \tag{2}$$

In the above equation (1), μ_B is the Bohr magneton, equal to 9.274×10^{-28} JG^{-1}, ΔW is the hyperfine energy splitting of a given particle, and W is the interaction

energy. A Landé g-factor may be defined to describe the change in the effective magnetic moment of an electron due to the magnetic dipole moment generated by a rotating charge. When only the electron is considered,

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \tag{3}$$

and when the interactions with the atomic nucleus are considered,

$$g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)} \tag{4}$$

For very low field strengths ($x << 1$), the interaction energy varies linearly with magnetic field, and can be calculated with a simplified version of equation (1).

$$W = g_F \mu_B F B \tag{5}$$

In the presence of an external magnetic field, the ground state of the atom will split into fine or hyperfine levels. When electrons at different ground states (hyperfine splitting) absorb photons of the same energy, they will move to higher energy hyperfine levels, but must obey a quantum mechanical selection rule based on angular momentum $\Delta F = 1$. However, atoms in a higher energy state have no such selection rule to obey when re-emitting photons as no polarization is favoured over the others, therefore all ΔF transitions are allowed. Through cycles of absorption and re-emission, the rubidium atoms will all end up in the highest angular momentum state with the highest F quantum number, but at ground level energy. Due to the angular momentum selection rule, these atoms are no longer able to absorb photons. This is the 'pumping' of atoms into their highest F state. However, if an oscillating magnetic field is applied at a special resonant frequency, the F quantum number in ground state atoms is again able to decrease, and optical pumping can occur again. This resonant frequency can be calculated with

$$\nu = \frac{g_F \mu_B B}{h} \tag{6}$$

Therefore, when an oscillating magnetic field is applied to the rubidium vapour cell with frequency not equal to this resonance frequency, atoms become 'locked-in' and cannot absorb photons from the discharge lamp, corresponding to high detector voltage. When this resonant frequency is achieved, the rubidium is able to continuously absorb incoming photons as it is depumped, and the detector voltage dramatically drops, referred to interchangeably as a 'dip' or 'peak' in this report.

In weak magnetic fields, the Landé g-factor of ^{87}Rb and ^{85}Rb in the rubidium vapour cell can be experimentally determined by measuring the resonance frequency and net magnetic field at which the resonant transition of each isotope is observed.

$$g_F = \frac{\nu h}{\mu_0 B} \tag{7}$$

The absorption of photons by a volume of gas with uniform gas density can be modelled by

$$I = I_0 e^{-\sigma_0 \rho l} \tag{8}$$

where I represents the outgoing flux of photons out of the gas, I_0 the incoming flux into the gas, l the path length through the gas, ρ is the gas density, and σ_0 is the maximum absorption cross-section. σ_0 is expected to be different from the geometrical cross-section calculated to represent of the geometrical diameter of the rubidium atom, which would be on the order of $(200 \times 10^{-12})^2$. These cross-section values represent the surface over which particles in the gas may absorb and re-emit photons in all directions. An appropriate fitting function for this relation would combine the cross-section and path length values to minimize fitting parameters.

$$f(x) = Ae^{bx} \tag{9}$$

The Zeeman Effect can be used to study the splitting of well-known atomic and molecular lines, and determine magnetic field strength at the origin of emission. This is a useful tool in astronomy, wherein stellar spectra are relatively easy measurements to achieve. Not only can the Zeeman effect in astronomy allow for magnetic activity of stars to be analyzed, but stellar conditions are a good laboratory to study Landé factors for both atoms and molecules in extreme environments [4]. Sunspots are cold and magnetically active areas of the sun's photosphere, which may be studied by the zeeman splitting of emission lines they create. This application is especially relevant as the number of sunspots increases during solar maximum, and our current solar cycle (cycle 25) is expected to peak next year. As described by laser manufacturer Lumispot, optical pumping is utilized in solid-state and semiconductor lasers to achieve high numbers of excited atoms, in order to ensure monochromatic emission [5].

Helmholtz coils are a set of coils that generate a magnetic field when supplied a current. Several sets of Helmholtz coils are utilized in this experiment, and the magnetic field generated by the horizontal sweep coils, in gauss is given by

$$\left|\vec{B}\right| = \frac{8.991 \times 10^{-3} NI}{R} \tag{10}$$

where N is the number of turns in each coil, R is the average coil radius, and I is the current through the coils in amps. The error of this field may also be calculated, assuming the error in N and R are zero.

$$\delta_{\vec{B}}^2 = \frac{8.991 \times 10^{-3} N}{R} \delta_I^2 \tag{11}$$

Experiment

This experiment was conducted with the TeachSpin Optical Pumping Experiment Apparatus, configured in two different ways for the Zeeman Effect and absorption sections of this experiment.

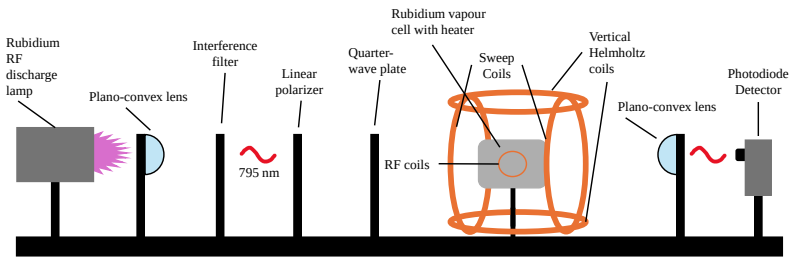

Figure 1: Optical track setup used for the Zeeman Effect sections of the optical pumping of rubidium experiment. Notable portions of this setup include the linear polarizer and quarter wave plate, which in combination, will circularly polarize the light that passes through them. Also note that the interference filter, which does not simply allow a single wavelength of light through, but a wavelength distribution that peaks around 800 nm. Plano-convex lenses are used to collimate light from the discharge lamp and rubidium cell.

In the linear and quadratic Zeeman Effect portions of this experiment, a linear polarizer and quarter-wave plate were used to (right-hand) circularly polarize rubidium resonance light from the radio-frequency (RF) discharge lamp. RF coils on the rubidium vapour cell were utilized to create an oscillating magnetic field, and both the sweep coils and main horizontal coils within the rubidium cell were used to generate horizontal fields perpendicular to the RF field. The vertical field Helmholtz coils were used to generate a vertical field equal and opposite to the vertical component of the Earth's magnetic field at the location of the experimental apparatus, and the track on which the apparatus laid was rotated to cancel the component of the field perpendicular to both the vertical and horizontal field components created by the pairs of Helmholtz coils. This initial 'zero-field' point was achieved using the magnetometer function of the iOS app Physics Toolbox, and was later confirmed by observing and minimizing the width of the zero-field transition on an oscilloscope.

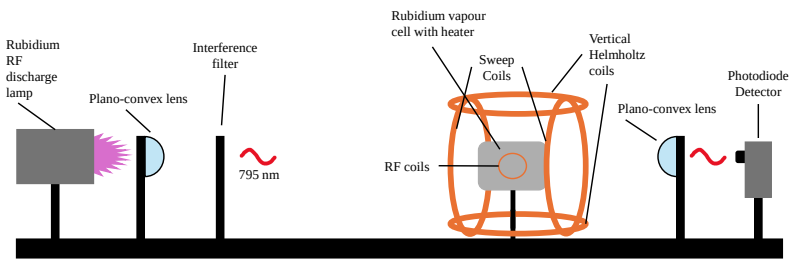

Figure 2: Optical track setup used in the absorption section of the optical pumping of rubidium experiment. This arrangement is very similar to that used in the Zeeman Effect portion of the experiment, with two components removed from the optical track.

For the absorption section of this experiment, the light was not polarized, and no horizontal or RF fields were necessary.

For the absorption section of this experiment, circular polarization, RF field oscillation, and the main horizontal field coils were all unnecessary. The linear polarizer and quarter wave plate were removed, so that no additional emission from the discharge lamp was filtered before entering the rubidium vapour cell. The RF coils and main horizontal coils were unplugged from the experimental controller so that current through those coils was guaranteed to be zero.

An Agilent DSO-X 2022A digital oscilloscope was used to visualize incident 795 nm photons on the photodiode detector. The save function on the oscilloscope was used to export CSV files of the response voltage as a function of time, which was then used in the analysis section of this report. An Agilent 34401A $6\frac{1}{2}$ digit digital multimeter was used to measure the voltage supplied to the main horizontal coils, as the 10-turn potentiometer did not supply the accuracy and precision necessary in this experiment. A Keysight 33210A 10 MHz Function/Arbitrary Waveform Generator was used to supply a sinusoidal voltage of set frequency and amplitude to the RF coils on the apparatus, generating an oscillating magnetic field.

The supplied optical pumping controller contains several sections, each dedicated to a different section of the experimental apparatus. The ten-turn potentiometer for the vertical field Helmholtz coils varies the supplied voltage from $0 - 1\,\mathrm{V}$. This voltage in Volts is numerically equal to the current through the coils in Amperes. The ten-turn potentiometer for the main horizontal field coils varies the supplied voltage from $0 - 1.5\,\mathrm{V}$, but hit a cutoff at about $0.5\,\mathrm{V}$ due to the fact that this experiment used the internal power supply rather than an external power supply for the main horizontal coils. The voltage was monitored

over a $2\,\Omega$ resistor, so current to the coils is numerically equal to twice the voltage. The ten-turn potentiometer for the sweep field Helmholtz coils range varies the supplied voltage from 0-1 V. The ten-turn potentiometer for the sweep field starting field varies the starting voltage from 0-1 V. These voltages are run through a $1\,\Omega$ resistor, and thus are numerically equal to the current supplied to the sweep Helmholtz coils.

The photodiode detector used in this experiment has three available 'gain resistance' settings: $10\,\mathrm{k}\Omega$, $1\,\mathrm{M}\Omega$, and $10\,\mathrm{M}\Omega$. However, the $10\,\mathrm{k}\Omega$ setting on the photodetector appeared to be non-functional, and was not used in the experiment. Therefore, the $1\,\mathrm{M}\Omega$ and $10\,\mathrm{M}\Omega$ settings were used to ensure the photodetector was not saturated, and did not show a flat maximum signal. In the absorption section of this experiment, both of the two available settings were investigated, in an attempt to gather data that did not overload the sensor.

KaleidaGraph v3.5 was used to locate the time at which each observed resonance occurred in comparison to the beginning of a given voltage sweep, allowing the instantaneous magnetic field to be calculated. The peak time was taken to be the time at which the lowest voltage was recorded, even though this may not necessarily match the real time at which the lowest voltage occurred. The consequences of this choice may be the source of some discontinuities discussed later.

Results and Analysis

Linear Zeeman Effect

A residual field calibration was conducted before low-level hyperfine splitting measurements were taken. After the optical rail on which the experiment was conducted was aligned with the Earth's magnetic field and the vertical component was minimized using the vertical Helmholtz coils, the zero-field transition was achieved with the sweep coils. The zero field transition occurred at a sweep coil current of $248(2)\,\mathrm{mA}$. The magnetic field generated by the horizontal sweep coils is given, in Gauss, by equation (10) where $N = 11$ turns, and $R = 0.1639\,\mathrm{m}$. The error in this residual field was calculated with equation (11), where δ_N and δ_R were assumed to be zero. The residual field for this section of the experiment was then found to be $0.150(2)\,\mathrm{G}$.

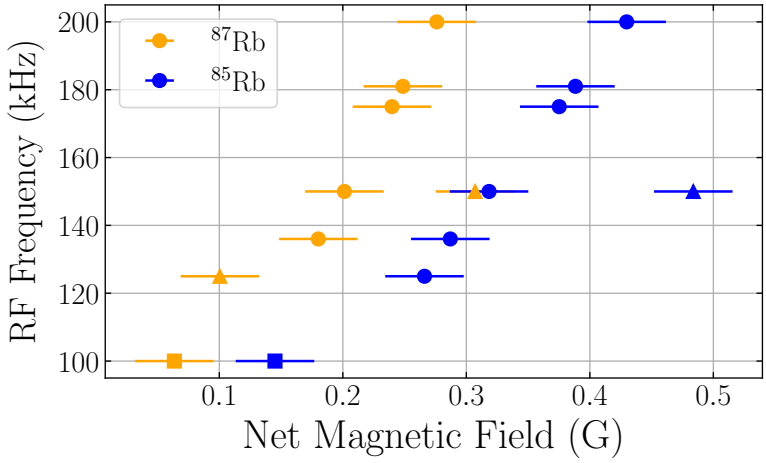

Figure 3: Resonance frequency of two rubidium isotopes in a weak magnetic field. For both ^{87}Rb and ^{85}Rb, the frequency varies linearly with magnetic field strength, though at different rates. The two isotopes are distinguished by marker colour, and marker shape is used to indicate data subsets used in the analysis. Vertical error bars are omitted as the RF frequency was held stable with a function generator. Horizontal error bars are a function of errors in both the residual field calibration as well as the generated sweep field, both of which are determined with equation (11).

The RF magnetic field was set to oscillate at a range of frequencies between 100 kHz and 200 kHz. For each sampled frequency, the 10-turn potentiometers on the optical pumping control panel were used to change voltage (and proportionally, current) across the sweep Helmholtz coils, thus sampling different magnetic field strengths. Several transitions were observed in which optical pumping occurred and the signal to the photodetector dropped. These dips indicated the point at which the fixed resonance frequency matched the resonance frequency of each isotope for a given magnetic field. With both the magnetic field and resonance frequency known, the Landé g-factor of each isotope of rubidium was calculated using equation (7).

Table 1: Landé g-factors for two isotopes of rubidium.

Data Set	g_F - ^{87}Rb	g_F - ^{85}Rb
Complete	0.62 ± 0.24	0.34 ± 0.07
Limited I	0.55 ± 0.15	0.32 ± 0.04
Limited II	0.53 ± 0.01	0.335 ± 0.002

The complete data subset contains points of all shapes as shown in figure 3. The Limited I subset of points excludes data with square markers, and the

limited II subset excludes both square and triangle markers. These adjusted data sets were created to allow different average g_F factors to be calculated by eliminating outliers to minimize uncertainty.

Overall, each of the data subsets result in calculated Landé g-factors similar to $g_F(^{87}\text{Rb}) = \frac{1}{2}$ and $g_F(^{85}\text{Rb}) = \frac{1}{3}$, with 5/6 values within error of the theoretical value.

This section of the experiment also required a calibration of the main horizontal field coils, so that they could be utilized in the following section to achieve much greater magnetic field strength and study hyperfine splitting. With the Landé g-factors of each isotope of rubidium now determined, they were used to calculate the magnetic field while resonance frequency was swept.

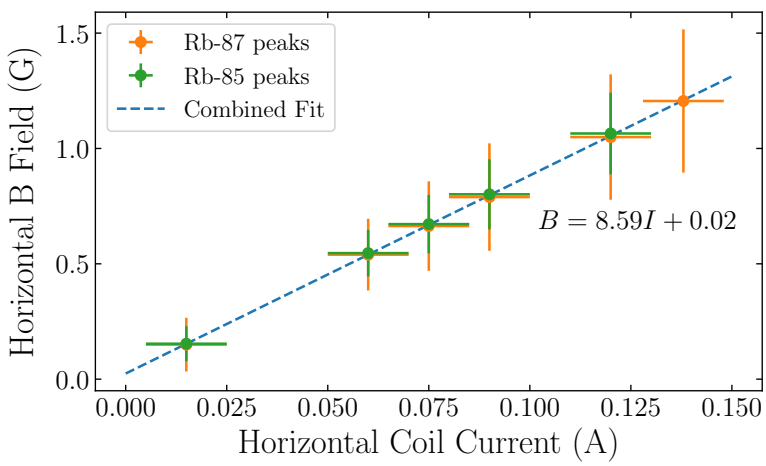

Figure 4: Determination of the field/amp conversion factor for the main horizontal coils. A conversion of $B = 8.59I + 0.02$ was calculated for the main horizontal field coils. The two rubidium isotopes should all fall along the same line, and do indeed find the same conversion within error, due to large error. Horizontal error bars representing coil current are very large due to the method with which the current measurements were taken. Rather than using a digital multimeter to take precise measurements, horizontal coil current was translated from the controller dial. Vertical error bars are a function of errors in the residual field, sweep field contribution, and calculated Landé g-factors for each isotope. The greater uncertainty in g_F - ^{87}Rb compared to g_F - ^{85}Rb is reflected in the figure.

For each sampled resonance frequency, a combination of the main horizontal coils and the horizontal sweep coils was used to find the peaks corresponding to each isotope of rubidium. The expected net magnetic field was calculated, and the residual field subtracted. The sweep coil contribution was subtracted, then the field contribution from the main horizontal coils was plotted as a function of main coil current.

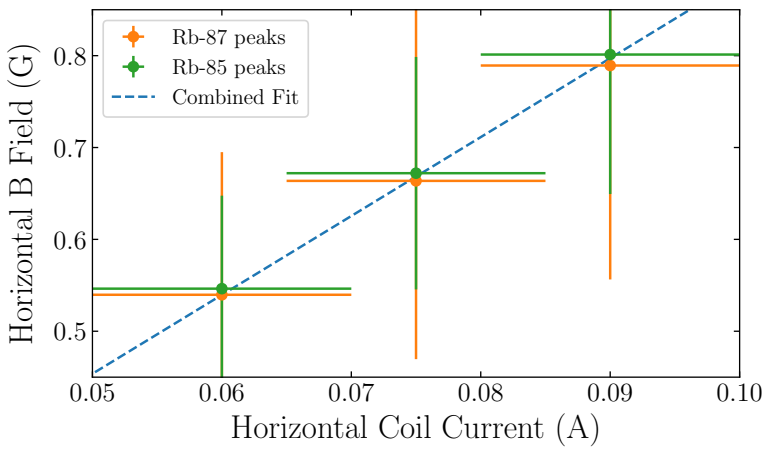

Figure 5: Enlarged section of figure 4, corresponding to the $\nu = 400 - 600\,\text{kHz}$ resonant frequency measurements. The two isotopes of rubidium clearly define their own unique horizontal coil conversions, an error which likely originates from the divergence of the calculated Landé g-factors from their theoretical values.

A fit performed only to the ^{87}Rb series found a conversion of $B = 8.27I + 0.01$, and the ^{85}Rb series found a conversion of $B = 9.14I + 0.04$. On page $13 - 4$ of the TeachSpin Optical Pumping of Rubidium Manual [1], the approximate current-to-field conversion for the main horizontal coils is given as $B = 8.8\,\text{GA}^{-1}$. While it is explicitly listed as only an approximation, the fact that the net effect of both isotopes approaches this value with a near zero gauss y-intercept indicates the conversion was well-behaved.

Quadratic Zeeman Effect

Next, the region in which interaction energy does not vary linearly with magnetic field was investigated. In this region, the hyperfine splitting energies are determined by equation (1). For both isotopes of rubidium, resonance is achieved at two different magnetic field strengths. Visible transitions and the field strength at which they occur are compared to the magnetic field predicted by equation 1 for a given resonance frequency.

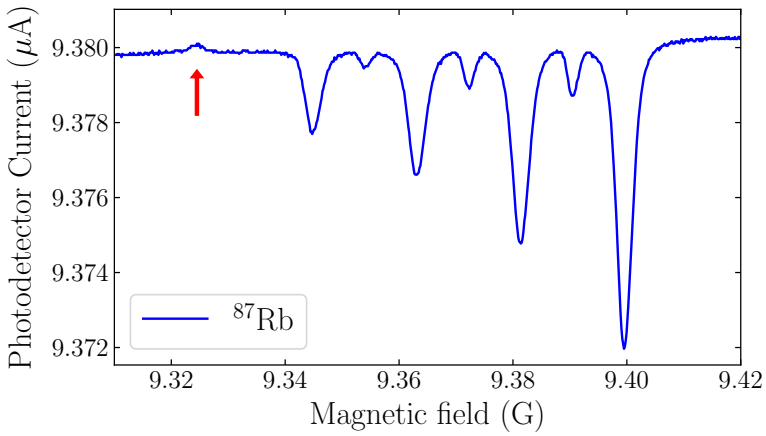

Figure 6: Spectrum of rubidium-87 at 6.4 MHz. Double-quantum transitions are visible but shallow, appearing between the larger quantum transitions. The first peak, in which gas opacity decreases rather than increasing, is marked with a red arrow. A second smaller peak is recorded in table 2, but not indicated here. Magnetic field refers to the net magnetic field, a sum of the residual field after achieving the narrowest possible zero field transition, the main horizontal field using the conversion factor determined in figure 4, and the field due to the instantaneous current through the sweep coils at the time the transition occurred. Voltage across the photodetector has been converted to current, which can be insightful when a power/amp rating for the photodetector is provided.

Figure 6 shows hyperfine splitting - the separation of energy levels previously degenerate at weaker magnetic fields. Dips in the spectrum appear when resonance is achieved for a particular quantum transition, or small change in energy level, and the gas is depumped such that the gas may begin absorbing photons again. The fact that dips 1-3 are viewed as peaks rather than dips could indicate a difference in the population of those particular energy levels compared to other transitions.

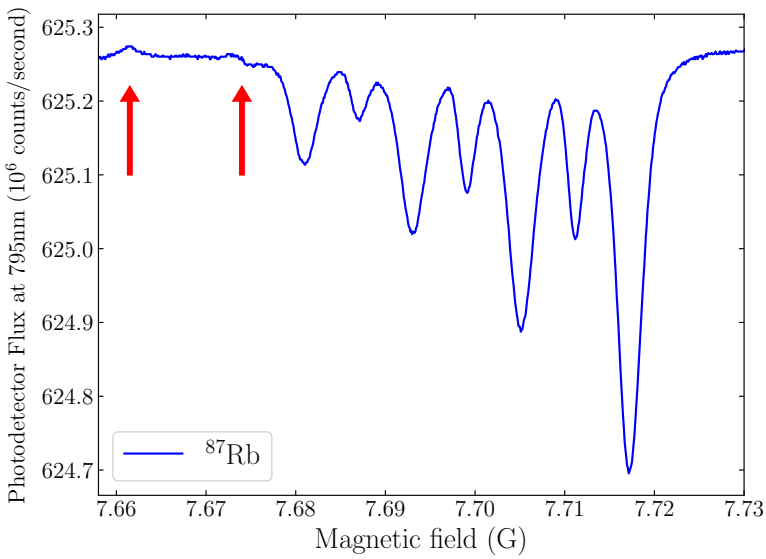

Figure 7: Spectrum of rubidium-87 at 5.222 MHz. Double-quantum transitions are visible and more prominent than at the higher frequency, due to the comparative RF power being higher. Two very small transitions are marked with red arrows, though three were observed and recorded in table 2. Optimal RF power was not achieved, thus transitions 1-3 are difficult to discern. Photodetector current was converted to 795 nm photon count per second in order to provide greater insight to the changing transmission of circularly polarized photons from the RF discharge lamp through the rubidium vapour cell.

A detector spectral response of $0.6\,\mathrm{AW}^{-1}$ at 795 nm was given on page 3-3 of the experiment manual [1]. Photodector current was converted to photon flux, where the number of photons falling on the detector each second is given by $p = 0.6I\frac{\lambda}{hc}$, where p is the photon count per second, h is the Planck constant, c is the speed of light in a vacuum, I is the photodetector current in Amperes, λ is the wavelength of the counted photons in metres, and $0.6\,\mathrm{AW}^{-1}$ is the spectral response of the diode at $\lambda = 795\,\mathrm{nm}$.

In the case of the two rubidium-87 spectra, each representative spectrum shows four highly prominent peaks, three lesser peaks, and some small extrema (some marked with arrows) at the low end of field strength. These marked transitions are unique in that they are much weaker, and represent the rubidium vapour opacity decreasing rather than increasing. The rubidium-85 response showed no such phenomenon in increased detection rather than decreased detection.

Table 2: Observed transitions in ^{87}Rb due to quadratic Zeeman splitting.

	DS1	DS2
Resonant Frequency (GHz)	5.222	6.400
RF Amplitude (mV_{PP})	200	200
Transition 1 $\vec{B}$ (G)	7.699(3)	9.353(3)
Transition 2 $\vec{B}$ (G)	7.705(3)	9.361(3)
Transition 3 $\vec{B}$ (G)	7.709(3)	Not Observed
Transition 4 $\vec{B}$ (G)	7.719(3)	9.372(3)
Transition 5 $\vec{B}$ (G)	7.725(3)	9.382(3)
Transition 6 $\vec{B}$ (G)	7.731(3)	9.391(3)
Transition 7 $\vec{B}$ (G)	7.737(3)	9.400(3)
Transition 8 $\vec{B}$ (G)	7.743(3)	9.409(3)
Transition 9 $\vec{B}$ (G)	7.749(3)	9.418(3)
Transition 10 $\vec{B}$ (G)	7.855(3)	9.427(3)

Ten transitions were observed out of an expected ten for $\nu = 5.222\,\text{MHz}$, while only nine were distinguishable for $\nu = 6.4\,\text{MHz}$. However, this table does not show the relative intensity of each transition, which minimizes the uncertainty with which the existence of the first three transitions are declared. The magnetic field strength at which each transition is achieved was calculated as a sum of the horizontal field due to the main coils, using the conversion found in figure 4, the horizontal field component due to the sweep coils and the instantaneous current through them at the time at which the transition was achieved, and the residual field component calculated after aligning the experimental apparatus that day with the Earth's magnetic field. Error on these magnetic field values are identical, due to identical uncertainties in time and current measurements, and the horizontal coil calibration.

Table 3: Comparison of observed to expected transitions for rubidium-87 in a magnetic field oscillating at 5.222 MHz.

	Expected B (G)	% Difference
Resonant Frequency (GHz)	5.222	
Transition 1 $\vec{B}$ (G)	7.439	3.51%
Transition 4 $\vec{B}$ (G)	7.456	3.52%
Transition 6 $\vec{B}$ (G)	7.497	3.52%
Transition 8 $\vec{B}$ (G)	7.479	3.53%
Transition 10 $\vec{B}$ (G)	7.490	3.53%

Table 4: Comparison of observed to expected transitions for rubidium-87 in a magnetic field oscillating at 6.4 MHz.

	Expected B (G)	% Difference
Resonant Frequency (GHz)	6.400	
Transition 1 $\vec{B}$ (G)	9.114	2.62%
Transition 4 $\vec{B}$ (G)	9.133	2.62%
Transition 6 $\vec{B}$ (G)	9.150	2.63%
Transition 8 $\vec{B}$ (G)	9.168	2.63%
Transition 10 $\vec{B}$ (G)	9.185	2.64%

Equation (1) was solved for each particular resonance frequency and allowed transition. These transitions were then compared to the observed single quantum transition, with the highest $\vec{B}$ transition assumed to correspond to transition 10. Then the percent difference between the expected and observed magnetic field for each transition was calculated, and is shown to be quite consistent in tables 3 and 4. Table 3 shows a systematic 3.5% departure from the expected magnetic field for each change in energy level, while table 4 shows a consistent 2.6% error. The largest contribution to the net magnetic field is from the main horizontal coils, so large and consistent departures from the expected field may represent a systematic error in using the calculated conversion factor. However, while these errors are present and nontrivial, they are small enough to not indicate any extreme errors in the procedure or solving of the Breit-Rabi equation.

Because rubidium-85 has a different atomic nucleus than rubidium-87, a different range of nuclear spin values are available, resulting in a different number of split energy levels at high magnetic field strengths. These unique hyperfine levels are shown through the transitions in figure 8.

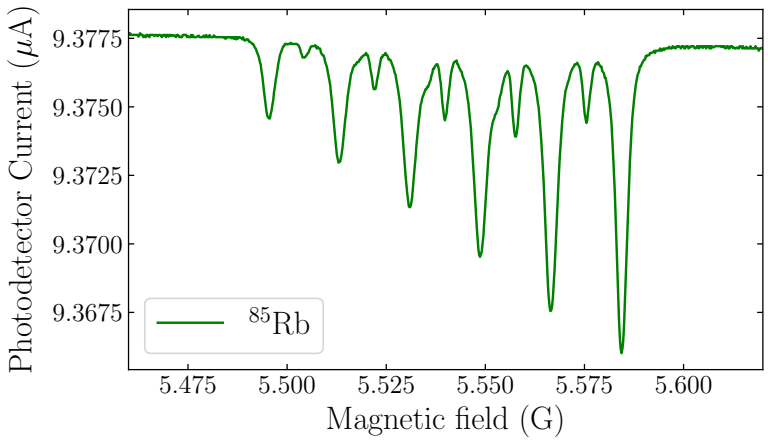

Figure 8: Spectrum of rubidium-85 with RF field frequency $\nu = 2.495\,\mathrm{MHz}$. Six single quantum transitions and five double quantum transitions clearly resolved in this spectrum. No peaks indicating increased gas transmission are visible in this spectrum, as each transition corresponds only to an increase in absorbed photons.

The net magnetic field strength at which each of these absorption lines occurs in figures 8 and 9 is summarized in table 5, wherein transition 1 corresponds to the first single quantum transition, and transition 11 corresponds to the sixth single quantum transition.

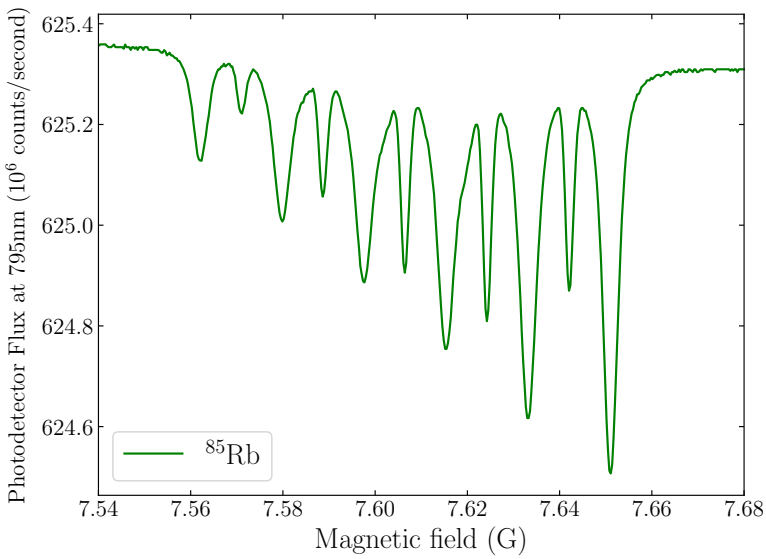

Figure 9: Spectrum of rubidium-85 with RF field frequency $\nu = 3.466\,\text{MHz}$. Photodetector current has been converted to counts per second of 795 nm photons, to better visualize the changing absorption of the rubidium gas within the cell. This spectrum utilizes a higher RF power (greater amplitude of oscillation set on the function generator) than figure 8, which manifests as more prominent double quantum transitions, and wider single quantum transitions.

Because the supplied RF power clearly has an effect on the intensity of transitions, different V_{PP} amplitudes on the function generator were investigated for rubidium-85. These various trials are showcased in table 5.

Table 5: Observed transitions in ^{85}Rb due to quadratic Zeeman splitting.

	DS3	DS4	DS5	DS6	DS7
Resonant Frequency (GHz)	3.466	3.466	3.466	2.495	2.495
RF Amplitude (mV_{PP})	200	200	150	150	100
Transition 1 $\vec{B}$ (G)	7.657(3)	7.658(3)	7.734(3)	5.612(3)	5.612(3)
Transition 2 $\vec{B}$ (G)	7.666(3)	7.667(3)	7.752(3)	5.616(3)	5.616(3)
Transition 3 $\vec{B}$ (G)	7.675(3)	7.675(3)	7.770(3)	5.620(3)	5.621(3)
Transition 4 $\vec{B}$ (G)	7.683(3)	7.684(3)	7.789(3)	5.625(3)	5.626(3)
Transition 5 $\vec{B}$ (G)	7.693(3)	7.693(3)	7.805(3)	5.630(3)	5.630(3)
Transition 6 $\vec{B}$ (G)	7.702(3)	7.702(3)	7.823(3)	5.634(3)	5.635(3)
Transition 7 $\vec{B}$ (G)	7.710(3)	7.711(3)	7.841(3)	5.639(3)	5.639(3)
Transition 8 $\vec{B}$ (G)	7.719(3)	7.720(3)	7.859(3)	5.644(3)	5.644(3)
Transition 9 $\vec{B}$ (G)	7.728(3)	7.729(3)	7.877(3)	5.648(3)	5.649(3)
Transition 10 $\vec{B}$ (G)	7.737(3)	7.738(3)	7.895(3)	5.653(3)	5.653(3)
Transition 11 $\vec{B}$ (G)	7.746(3)	7.747(3)	7.912(3)	5.658(3)	5.658(3)

DS3-7 are labels for 'data set' 3-7, used to differentiate absorption spectra measured at the same RF frequency and possibly amplitude. The magnetic field strength at which each transition is achieved was calculated as a sum of the horizontal field due to the main coils, using the conversion found in figure 4, the horizontal field component due to the sweep coils and the instantaneous current through them at the time at which the transition was achieved, and the residual field component calculated after aligning the experimental apparatus that day with the Earth's magnetic field. Error on these magnetic field values are identical, due to identical uncertainties in time and current measurements, and the horizontal coil calibration. Small differences present themselves in the magnetic field for a single resonant frequency when different RF powers are considered. This may be in part due to the fact that the absorption signature is not symmetric (nicely visible in figure 8), but the transition field was calculated for the lowest recorded photodiode voltage.

Table 6: Comparison of observed to expected transitions for rubidium-85 in a magnetic field oscillating at 3.466 MHz

	Expected B (G)	% Difference	% Difference	% Difference
Data Set		DS3	DS4	DS5
Transition 1 $\vec{B}$ (G)	7.392	3.58%	3.59%	4.63%
Transition 3 $\vec{B}$ (G)	7.396	3.65%	3.66%	4.81%
Transition 4 $\vec{B}$ (G)	7.409	3.59%	3.59%	4.87%
Transition 5 $\vec{B}$ (G)	7.413	3.65%	3.66%	5.05%
Transition 6 $\vec{B}$ (G)	7.426	3.59%	3.60%	5.10%
Transition 7 $\vec{B}$ (G)	7.430	3.65%	3.66%	5.29%
Transition 8 $\vec{B}$ (G)	7.443	3.59%	3.60%	5.34%
Transition 9 $\vec{B}$ (G)	7.447	3.66%	3.67%	5.53%
Transition 10 $\vec{B}$ (G)	7.460	3.59%	3.60%	5.58%
Transition 11 $\vec{B}$ (G)	7.477	3.47%	3.49%	5.58%

While eleven absorption peaks are clearly visible in both figures 8 and 9, only ten transitions and thus peaks are predicted by equation (1). Because of that, transition two was assumed to be the the transition not predicted by these calculations as it was the smallest in amplitude. The remaining transitions were then compared to the expected magnitude of electric field to which they should be observed.

Table 7: Comparison of observed to expected transitions for rubidium-85 in a magnetic field oscillating at 3.466 MHz.

	Expected B (G)	% Difference	% Difference
Data Set		DS6	DS7
Transition 1 $\vec{B}$ (G)	7.392	5.28%	5.29%
Transition 3 $\vec{B}$ (G)	7.396	5.38%	5.39%
Transition 4 $\vec{B}$ (G)	7.409	5.28%	5.29%
Transition 5 $\vec{B}$ (G)	7.413	5.38%	5.39%
Transition 6 $\vec{B}$ (G)	7.426	5.28%	5.29%
Transition 7 $\vec{B}$ (G)	7.430	5.38%	5.39%
Transition 8 $\vec{B}$ (G)	7.443	5.28%	5.29%
Transition 9 $\vec{B}$ (G)	7.447	5.38%	5.39%
Transition 10 $\vec{B}$ (G)	7.460	5.28%	5.29%
Transition 11 $\vec{B}$ (G)	7.477	5.19%	5.20%

While in the case of rubidium-87, the difference between the calculated and observed magnetic field for each transition at a fixed resonance frequency remained constant over each transition, both trials for rubidium-85 show alterna-

tion between high and low values. However, these do not necessarily correspond to if the transition is single or double quantum, and the final absorption peak shows a drastically lower difference from its predicted postition in each data set. Tables 6 and 7 also show a correlation between lower RF amplitude being applied to the RF coils, and a higher percent difference from the expected magnetic field strength for which an absorption feature should appear.

Absorption of Rb resonance radiation by atomic Rb

Table 8: Temperature dependence of rubidium vapour absorption of infrared photons with a photodiode detector gain resistance = 1 MΩ.

Rb Cell Temperature (C)	Detector Voltage (V)
29.9(1)	1.9916(5)
39.9(1)	1.9955(4)
49.9(1)	2.749(5)
59.9(1)	2.765(2)
70.0(1)	2.757(7)
79.9(1)	-1.810(2)
89.9(1)	-3.971(2)
99.9(1)	-5.925(1)
109.9(1)	-6.9(1)
119.9(1)	-7.441(2)
130.0(1)	-5.435(2)

Detector amplifier gain and DC offset were not held fixed for each data point collected, so the data in this table is not entirely reflective of the changing absorption of rubidium vapour in the cell. When the amplified voltage is converted to real detector voltage, and cell temperature to atomic density according to a conversion generated with data from page 4-2 of the experimental manual [1], the pattern by which absorption changes is revealed.

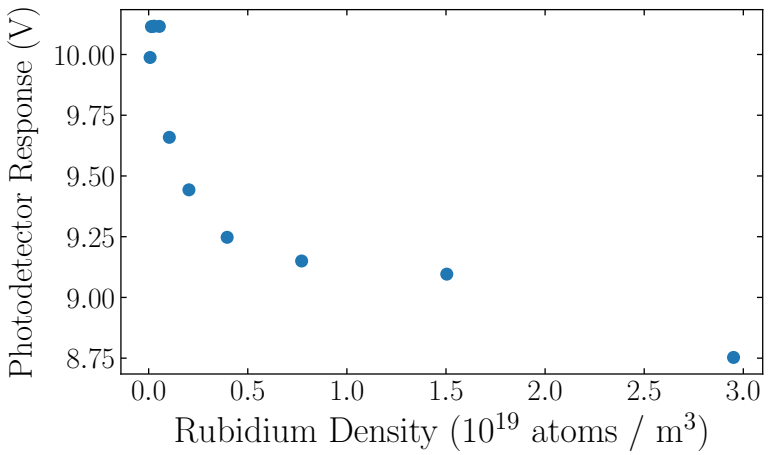

Figure 10: Temperature dependence of absorption of rubidium gas in a xenon buffer gas. The intensity of 795 nm photons hitting the photodiode detector resulted in a voltage that could be amplified and recorded on a multimeter. At low gas density, the detector is saturated, and the detector response is flat until it falls below the threshold at which changes in intensity can be recorded again. Both horizontal and vertical error bars are included in this figure, but fluctuations in voltage were extremely small due to the gain used to observe it, and the estimated error in the provided conversion for gas density was assumed to be near zero.

In order to be fit to an exponential function of the form given in equation (9), some data points had to be excluded, and the voltage shifted toward 0. The first 3 data points, corresponding to a saturated photodetector, were excluded. Additionally, the final data point does not follow the expected behaviour that the opacity of the vapourized rubidium stabilizes at high enough gas density. In order to achieve a reasonable fit and estimate the absorption cross-section, this point was also excluded. The reduced data set and corresponding fit are shown in figure 11.

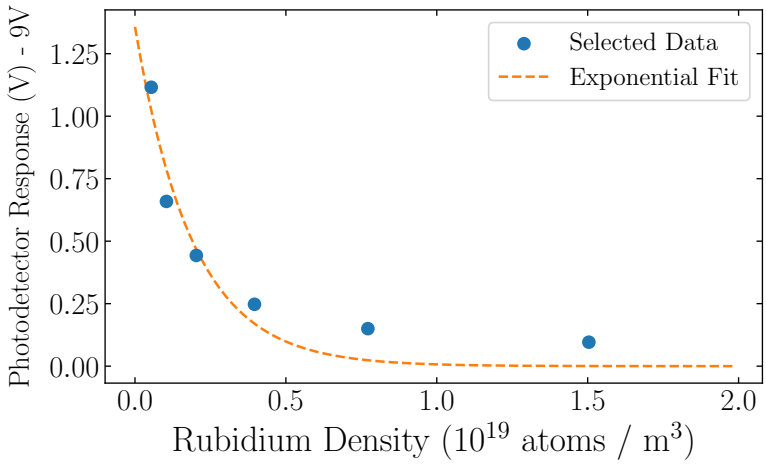

Figure 11: A refined absorption response of vapourized rubidium in xenon due to changing temperature. Temperature varies proportionally with the rubidium gas density, and a decreased photodector voltage corresponds to greater gas absorption. The selected data was fit to a function of the form shown in equation (9), and found to be $V = 1.4(2)exp[-5(1)\rho]$. Both horizontal and vertical error bars are included in this figure, but are too small to be visible.

According to the fit in figure 11,

$$-\sigma_0 \rho l = (-5.2 \pm 1.4)\rho$$

On page 3-10 of the optical pumping experiment manual [1], the rubidium cell outer length is given as 36 mm with a wall thickness of 1.5 mm. Given this information, the path length through the gas is 0.033 m. The maximum absorption cross section can then be calculated by comparing the arguments of the exponential in equation (8) to the fit achieved in figure 11.

$$-0.033\sigma_0 \times 10^{19} = -5.2 \pm 1.4$$

$$\sigma_0 = (1.6 \pm 0.4) \times 10^{-17}\mathrm{m}^2$$

The geometrical cross-section of a rubidium atom with a radius of about 2×10^{-12} m is on the order of $4 \times 10^{-20}\,\mathrm{m}^2$.

Because the detector preamplifier was saturated at low temperatures in this trial, resulting in a flat response, the measurements were repeated with the detector switched to the 10 MΩ setting. The detector 10 kΩ setting was not operational, so this was the only option for troubleshooting.

Table 9: Temperature dependence of rubidium vapour absorption of infrared photons with a photodiode detector gain resistance = 10 MΩ.

Rb Cell Temperature (C)	Detector Voltage (V)
30.0(1)	6.626(2)
39.9(1)	6.647(1)
50.0(1)	6.6669(1)
59.9(1)	6.686(1)
70.0(1)	6.701(1)
80.0(1)	6.668(1)
89.9(1)	6.662(1)
100.0(1)	6.660(1)
110.1(1)	6.636(2)

With the detector preamplifier set to the 10 MΩ gain resistance, the amplifier gain and DC offset were able to stay constant over the whole temperature range, as the preamplifier was saturated and showed virtually no response over the entire range. Because the response was clearly flat, measurements were ended early at $110°C$. This data could not be analyzed, but is included as proof of attempted troubleshooting after the previous results (and as a testament to the time commitment of this experiment).

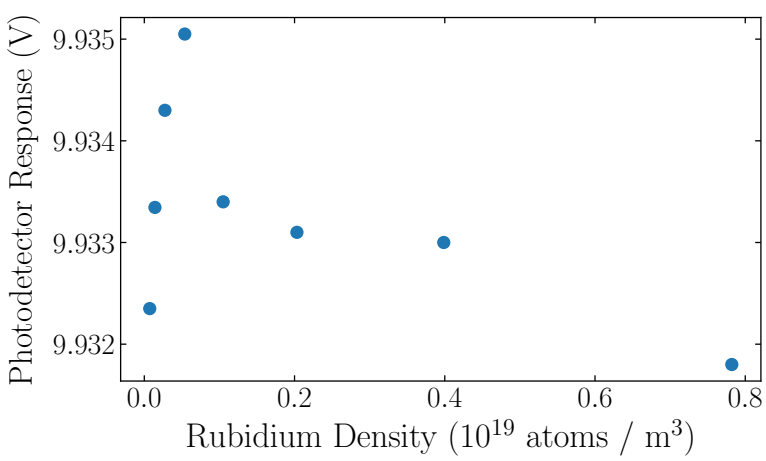

Figure 12: Absorption of rubidium as a function of cell temperature with the photodiode detector set to its 10 MΩ setting. Lower temperatures were achieved in this trial, thus lower atomic rubidium densities. There is no coherent exponential decay to the detector voltage, indicating no increase in photon absorption. However, the absorption signature was visible in the previous trial, so this flat response is more likely due to saturation of the detector rather than any exotic behaviour of the rubidium gas.

Discussion

Linear Zeeman Effect

The best Landé g-factor for rubidium-87 calculated in this experiment was $g_F = 0.55 \pm 0.15$, which is within error of the expected value of $g_F = 1/2$. The best Landé g-factor for rubidium-85 calculated in this experiment was $g_F = 0.335 \pm 0.002$, which is within error of the expected value of $g_F = 1/3$. Each of these 'best' factors was calculated with an average of g-factors calculated according to equation 7 for various resonant frequencies, and were achieved by constraining the data that made these averages. This section of the experiment was the first section completed, thus these measurements were made whilst still becoming familiar with the experimental apparatus. As such, data was collected twice for $\nu = 150\,\mathrm{kHz}$. In one of these collections, the transition of each isotope was recorded separately, and in the other, both the isotope absorption features were visible within one sweep of the horizontal sweep coils. These should have resulted in equal magnetic field calculations, but did not. This is most likely due to an error in the recording of the sweep field settings, as the voltage range recorded for the single-response trials is higher than should be possible whilst only observing one peak. Therefore, this larger range resulted in a larger net magnetic field being calculated for those two trials, and the points must be excluded in order to obtain a more accurate Landé g-factor. Due to the calculation of the g-factors as an average of individually determined values rather than a graphical fit with slope analysis, the uncertainty reported in table 1 is representative of the standard deviation of these values rather than a fitting error, so limiting the included data greatly reduces the standard deviation, and thus error. This does not necessarily reflect that the value is narrowing on a more accurate value, just that the various g_F are in better agreement.

In the calibration of the horizontal field coils, the calculated g_F values were used rather than the theoretical values, which only appeared to cause very small errors in this section of the experiment (may be the source of larger errors later on). If the previous constants had been calculated accurately, the two isotopes would all lie along a single conversion factor line, but the two series define slightly different slopes. However, this difference is very small within the regime investigated in figure 4, and the errors associated with each data point are much more significant than the difference in conversion factors. A fit performed to the ^{87}Rb series found a conversion of $B = 8.27I + 0.01$, and the ^{85}Rb series found a conversion of $B = 9.14I + 0.04$. On page $13 - 4$ of the TeachSpin Optical Pumping of Rubidium Manual [1], the approximate current-to-field conversion for the main horizontal coils is given as $B = 8.8\,\mathrm{GA}^{-1}$. While it is explicitly listed as only an approximation, it confirms the theory that error in this conversion factor is linked to the Landé g-factor, as the isotope with a smaller percent difference from the expected value for g_F has a smaller percent difference from the expected value for the horizontal field coil calibration. The y-intercept of each calibration, most notably the combined fit, is near zero. This corresponds to zero magnetic field contribution from the main horizontal

coils when no current is run through them, which is physical and expected of Helmholtz coils. However, when these coils were plugged into the optical pumping controller with the potentiometer turned to 0.00 V, a small nonzero voltage was measured with a digital multimeter. This could be the origin of the small nonzero y-intercept B_0. The agreement between the fits and this expected intercept may be an indicator that the conversion factor is highly accurate as shown in figure 4, even with the propogated errors of using calculated Landé g-factors.

Quadratic Zeeman Effect

Though many trials were conducted wherein the frequency and amplitude of the signal supplied to the RF coils were varied, the observed spectra of each isotope were always extremely similar to those shown in the analysis. This is problematic due to the disagreement between the calculated and observed number of absorption lines due to energy level transitions, as well as the fact that some lines indicate emission for rubidium-87. If optimal RF power had been achieved, these first three small signatures in the rubidium-87 spectrum would have been much more prominent, and could have been investigated more thoroughly. In the state they are, these transitions are extremely weak, and their existence is questionable in some cases.

The confusion associated with these weak spectral signatures created difficulty in matching the theoretically determined resonant peaks to those observed. Double quantum transitions were not considered to be these transitions in the case of rubidium-87, which forced the small transitions 1 and 3 to be matched to the first two calculated transitions. Though tables 3 and 4 show that this was likely a correct assumption as the percent difference appears systematic, it highlights the need for observations to be conducted at ideal RF power. Additionally, Landé g-factors should again be considered when presented with this systematic difference between the calculated and observed net field strength at which transitions occurred. Because the magnetic field strength in this regime is dominated by the contribution from the main horizontal coils, error in observed magnetic field is closely tied to the error in this component. While the conversion factor showed very little impact due to the different slopes in the $0 - 1.0$ G range in figure 4, these differences would be exaggerated at the high magnetic field strengths used here. Errors introduced by calculating a conversion factor based on calculated g_F values would compound, and present here as a consistent error in each value of magnetic field. This may be the origin of the systematic error between the expected and observed resonances of rubidium-87.

In the case of rubidium-85, both the single and double quantum transitions are clearly resolved, and each show photon absorption. 11 absorption signatures are visible, but theory predicts 10 peaks. Because only 6 single quantum transitions are present, the transitions calculated using equation (1) must include a subset of the double quantum transitions as well. However, the analytical choice to compare calculated values to transitions 1 and 3-11 clearly had drawbacks, as the percent difference between calculated and observed fields for each transition

no longer appeared systematic. These values oscillated between a low and high percent difference for transitions 1 and 3-10, with transition 11 showing a much smaller departure from the theoretical value. This low-high pattern does not align with the single-double quantum transition pattern of the spectrum, and the fact that transition 11 greatly differs from the other 9 transitions does not align well with the fact that the skipped transition was at the low end of the spectrum rather than the high end. While these differences between calculated and observed transitions do vary, the degree by which they deviate is very small. This implies that like with rubidium-87, some systematic error is present in the determination of magnetic field strength for each transition.

Small differences present themselves in the magnetic field for a single resonant frequency when different RF powers are considered, a phenomenon visible in table 5. This may be in part due to the fact that the absorption signature is not symmetric (nicely visible in figure 8), with a greater slope on the low field side of the transition, and lesser slope on the increasing field. Transition field was calculated at the point with the lowest recorded photodiode voltage. This creates a small skew toward lower magnetic fields of strength that increases with RF power, which matches the direction of magnetic field shift between data sets 3, 4, and 5 in tables 5 and 6. This asymmetry could be due to the mechanism by which the resonance dip occurs. When the oscillating magnetic field provides an energy that is equal to the energy difference of a given transition, atoms that have been pumped to their highest angular momentum state may shift downward, allowing them to resume absorbing photons. This continues as the magnetic field that corresponds to this swept past. As the magnetic field increases, the oscillating field no longer has the energy to change the angular momentum state of the atoms, and they are again pumped to the highest state. If some portion of these atoms had been transported to the lowest possible state of angular momentum, this pumping mechanism would have to be applied multiple times to restore the gas to the highest momentum, non-absorbing state. It is possible that the emission peak asymmetry originates from the time difference between all atoms moving into an absorbing state, and being pumped back to a non-absorbing state.

All uncertainties in this section of the experiment were under-estimated. The uncertainty in each magnetic field strength was calculated based on uncertainty of the time at which the peak was measured on the oscilloscope, as well as errors in the residual field and horizontal field. As previously discussed, the main horizontal field coil calibration may have been a larger source of error than was explored when it was calculated. The choice to select the time of the transition occurring as the time corresponding to the lowest recorded photodiode voltage should have also introduced an error due to the fact that the oscilloscope may not have had a high enough sample rate to capture this exact instant.

Absorption of Rubidium resonance radiation by atomic rubidium

The maximum absorption cross-section of rubidium was found to be $\sigma_0 = (1.6 \pm 0.4) \times 10^{-17}$ m^2. This is three orders of magnitude larger than the geometrical cross section of about $\sigma = 4 \times 10^{-20}$ m^2. These results can imply that if the particle theory of light is considered, the rubidium atom is capable of interacting with a particle that is considerably far outside its atomic radius. This is, however, an upper limit to the absorption cross-section, and the fact that it is indeed larger than the geometrical cross-section implies there were no errors in calculation.

However, the data used to calculate this value had to be greatly refined, and reflects large uncertainties due to the small sample size. For a larger gain resistance, voltage across the photodiode increases. Therefore attempting a measurement of the absorption of rubidium as a function of cell temperature with the photodetector set to the $10\,\mathrm{k}\Omega$ setting would likely result in a data set with a greater number of useable points, that could be fit to a greater accuracy and may yield a maximum absorption cross-section that is orders of magnitude more similar to the geometrical cross-section of the rubidium atom.

Conclusions

Optical pumping of rubidium was used to investigate the Zeeman Effect in both linear and quadratic regimes, allowing both fine and hyperfine splitting to be observed. The Landé g-factors of ^{87}Rb and ^{85}Rb were found to be 0.55 ± 0.15 and 0.32 ± 0.04, respectively, matching the theoretical values of $\frac{1}{2}$ and $\frac{1}{3}$ within error. Using the experimentally determined Landé g-factors of rubidium-87 and rubidium-85 to describe resonant transitions in a magnetic field, a conversion factor for the magnetic field (in gauss) generated by current through the apparatus' main horizontal coils was found to be $B = 8.59(5)I + 0.024(4)$. The hyperfine splitting of rubidium was observed and compared to the corresponding theoretical field strength, where it was found that a systemic error caused a small percent difference between the theoretical and observed fields. This systemic error was explored to possibly be due to the calculated horizontal field, absorption line asymmetry, or faulty identification of transitions. The absorption cross-section upper limit for rubidium was calculated to be $\sigma_0 = (1.6 \pm 0.4) \times 10^{-17}\mathrm{m}^2$, which appropriately is larger than the geometrical cross-section of the rubidium atom. Calculated values in each section of this experiment could be determined with higher accuracy if repeated with greater care in reducing the propogation of errors.

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